

FOUC

Proof: - We define

$$x(\alpha) = x^* + \alpha d \in \Omega$$

then $x(0) = x^*$. Now define the composite function $\phi(\alpha) = f(x(\alpha))$.

Then by Taylor's theorem

$$\underline{f(x^* + \alpha d) - f(x^*)}$$

$$= \phi(\alpha) - \phi(0)$$

$$= \phi'(\alpha) \alpha + o(\alpha)$$

$$= \alpha \nabla f(x^*) d^T + o(\alpha)$$

where $\alpha > 0$. Thus if $\phi(\alpha) > \phi(0)$ that is $f(x^* + \alpha d) > f(x^*)$ for small values of $\alpha > 0$,

$$\text{then } d^T \nabla f(x^*) > 0$$

— X —

$$\begin{aligned} \phi(\alpha) &= f(x(\alpha)) \\ &= f(x^* + \alpha d) \end{aligned}$$

$$\begin{aligned} \therefore \phi'(\alpha) &= \frac{d}{d\alpha} f(x^* + \alpha d) \\ &= \nabla f(x^*) d^T \end{aligned}$$

Page-66: $o(g(x))$; little oh of $g(x)$ represents a function that goes to zero faster than $g(x)$.

$$\alpha > 0$$

$$\Rightarrow \phi(\alpha) > \phi(0)$$

$$\Rightarrow f(x(\alpha)) > f(x(0))$$

$$\Rightarrow f(x^* + \alpha d) > f(x^*)$$

$$\Rightarrow \underline{f(x^* + \alpha d) - f(x^*)} > 0$$

$$\Rightarrow \alpha \nabla f(x^*) d^T > 0$$

$$\Rightarrow \nabla f(x^*) d^T > 0$$

$$\Rightarrow \nabla f(x^*) > 0$$